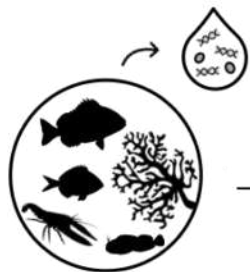


eDNA quality control

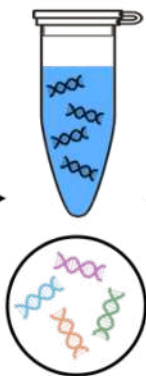
Emilie Boulanger
DNA-derived data management course
Santa Marta, Colombia
22/07/2026

Quality control Along the workflow from field to data

Field sampling



DNA extraction
& amplification



High-throughput
sequencing

```
CTCCAAAT  CCCCTACC
              CCCCGAAC
    CCCCGAAC  CCGCTAGT
    CCCCTACC  CTCCAAAT
CTCCAAAT  CCCCGAAC
    CCGCTAGT  CCCCTACC
    CCGCTAGT  CCGCTAGT
    CCCCGAAC  CCCCTACC
    CCCCTACC  CCGCTAGT
    CCGCTAGT  CTCCAAAT
    CCGCTAGT  CCCCGAAC
CTCCAAAT  CCGCTAGT
```

Bio-informatic
treatment

```
CTCCAAAT
CTCCAAAT
CTCCAAAT

CCCCGAAC
CCCCGAAC
CCCCGAAC

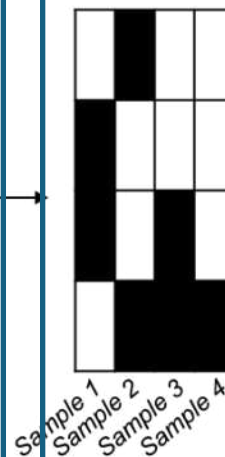
CCCCTACC
CCCCTACC
CCCCTACC

CCGCTAGT
CCGCTAGT
CCGCTAGT
```

Taxonomic
assignment



Ecological
dataset



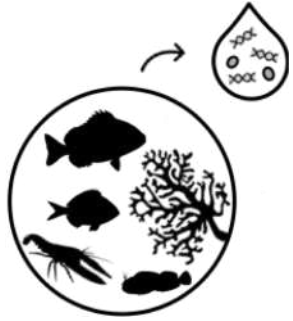
Emilie

Pieter

Elizabeth

Field sampling

Replicates and controls



Not much QC to do yet. But make sure to:

- Collect biological replicates (replicate seawater, river water, sediment, ... samples)
 - To minimize false negatives
 - Minimum 3
 - Ideally 6 or more Stauffer et al. 2021
- Collect and filter field blanks: miliQ water
 - To detect & filter false positives

Reminder

False positives – false negatives

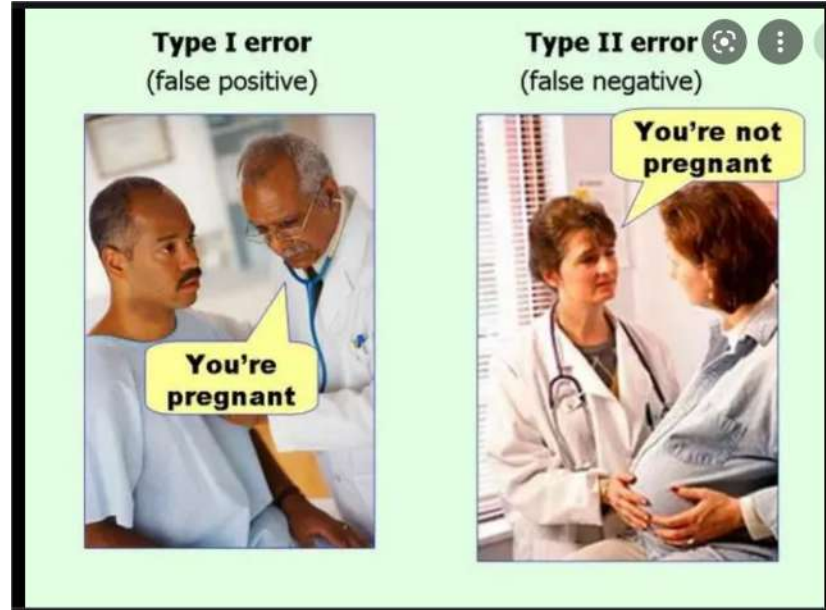
False positive

A species is detected although it is not present



In eDNA, can come from

- Contamination
- Amplification error
- Chimera
- Sequencing error
- Tag jumping
- Wrong assignments



False negative

A species is not detected although it is present



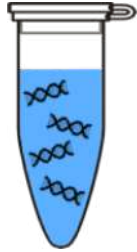
In eDNA, can come from

- Primer bias
- PCR inhibitors
- Undersampling
- Missing references

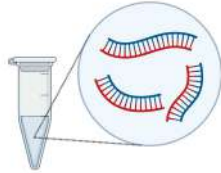
DNA extraction

Quantification

- Include extraction blanks
- Is there sufficient DNA for PCR?
- Is your DNA product contaminant-free?



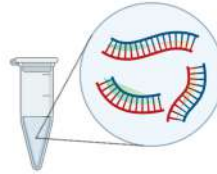
Quantification of DNA with the Nanospectrophotometer



- Spectrophotometer
- DNA concentration
- Potential RNA and protein contamination.



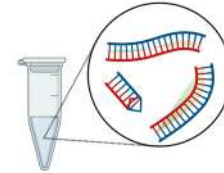
Quantification of DNA with the Qubit Fluorometer



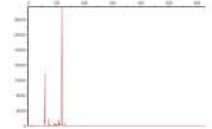
- Fluorometer
- Accurate DNA concentration
- High sensitivity with fluorescent dye



Quantification and Size of DNA with the TapeStation Gel Electrophoresis System

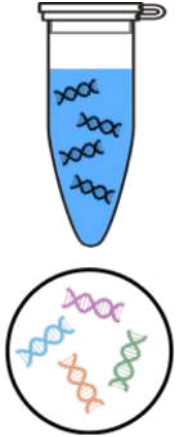


- Gel electrophoresis
- Accurate DNA sizing
- High sensitivity with fluorescent dye

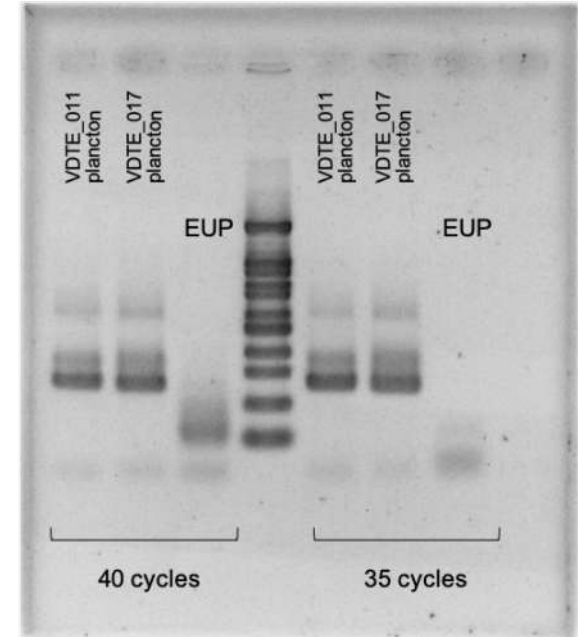
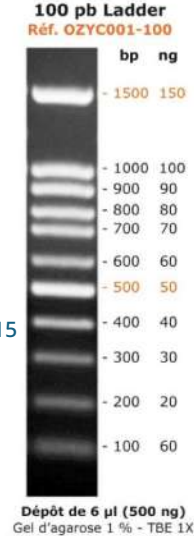


Amplification – PCR

Gel electrophoresis



- Include PCR blanks
- Amplify PCR replicates (technical replicates)
 - Minimum 3, ideally 8-12 Ficetola et al. 2015
- Is your amplicon the right size?
- Are your PCR blanks empty?



Sequencing & bioinformatics

FASTQ

- Sufficient read count per sample?
- Look within your fastq file at base quality score



FASTQ file sample:

```
@SRR6407486.1 1 length=100  
CCTCGTCTACAGCGACAACGCCGGAACGGGTGATGCGGGCCCTGGGCAACGGTTGACCCGGATCTGCCGATTGACCTACGTCGAAGTG  
+SRR6407486.1 1 length=100  
BBBBBFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
```

Sequence name
@SRR6407486.1 1 length=100

DNA sequence
CCTCGTCTACAGCGACAAC ... GATTGACCTACGTCGAAGTG

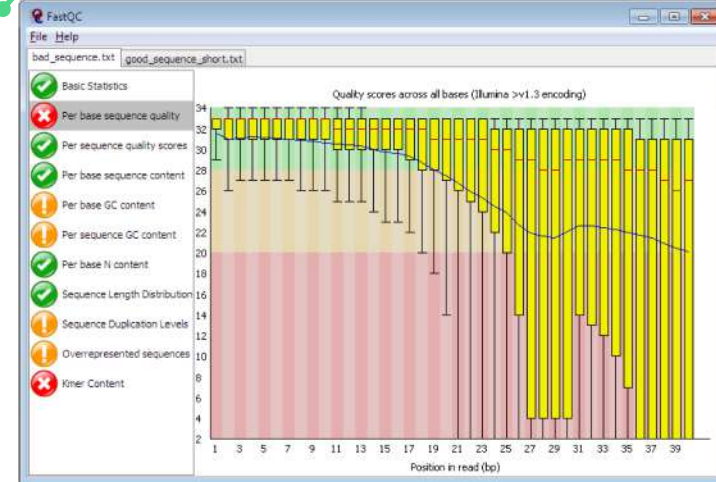
Quality line break
+SRR6407486.1 1 length=100

Quality scores
BBBBBFF

Base: T
Quality: 7

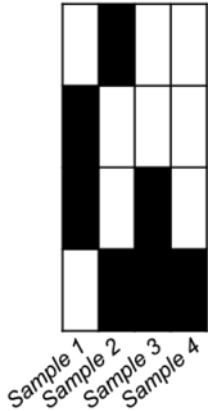
probability that this particular base was called incorrectly (log scale)

- Trim off low quality ends
- Filter whole reads
- Integrate into error model
- Report % of reads passing quality filter



Ecological dataset

Negative controls



- Curate taxonomic assignments
- Compare your true samples with negative controls
- Subtract contaminants from samples (either full or abundance-based)
- Filter on minimum read-count or relative-abundance threshold

